Data Ingestion

* Set of methods used to collect, clean, transform & organize raw data into meaningful insights & usable info.

Step 1:- Data sources.
Ex:- IaaS, SaaS, PaaS.

Step 2:- Data Ingestion.
Ex:- Real-Time processed, Batch-proces
data.

Step 3:- Data Processing.
Ex:- ETL

Step 4:- Data Storage
Ex:- Data ~~Storage~~ Lakes, Data Warehous

Step 5:- Data Users.

Foundational Layer :- Data Governance, Orchestratio
Monitoring.

CI/CD => Continuous Integration & Continuou
Development

Types of Data Ingestion

1) Batch Ingestion
* Collects data over a period of time & processes it together.

2) Real-Time Ingestion
* Collects data almost immediately after it is generated.

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3) Streaming Ingestion

* Continuously processes a flow of data as events arrive.

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Data Ingestion

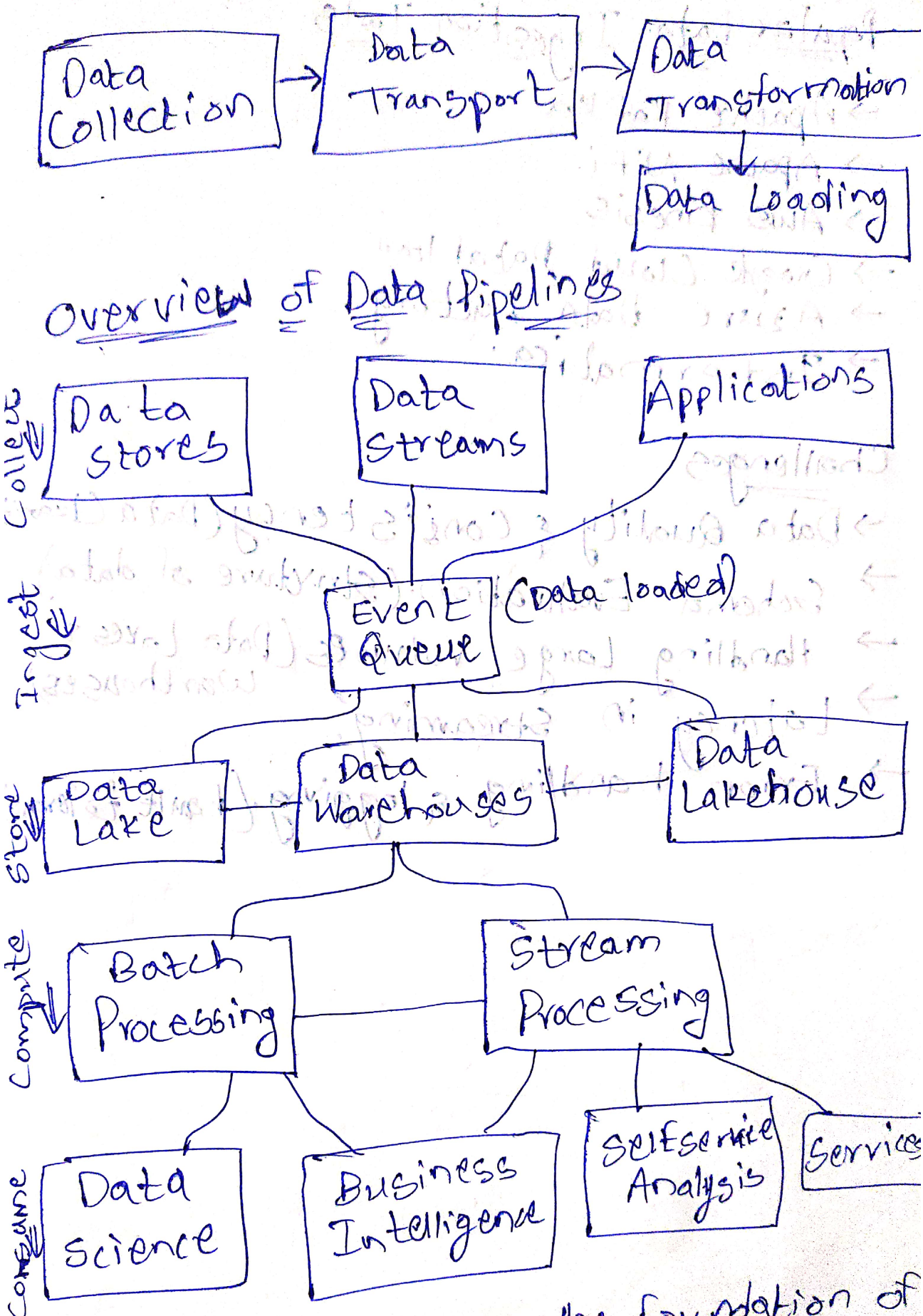
Ex:- Online Shopping Website

Customer Places Order
↓
Shopping application
↓
Data Ingestion
↓
Data Lake / Warehouse
↓
Sales Dashboard

Data Ingestion Pipeline Architecture

→ Source Systems ⇒ Data Producers.
⇒ Ingestion Layer ⇒ Tools & scripts to fetch data.
→ Storage Layer ⇒ Data Lake or warehouse
→ Processing Layer ⇒ ETL/ELT Transformations.
→ Consumption Layer ⇒ Analytics, ML & reporting.

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* Data ingestion is the foundation of any data engineering process.

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Popular Data Ingestion Tools

→ Apache Kafka
→ Apache Nifi
→ AWS Kinesis
→ Google Cloud Dataflow
→ Azure Data Factory
→ Informatica.

Challenges

→ Data Quality & Consistency (Data Cleaning)
→ Schema Evaluation. (Structure of data)
→ Handling Large Volumes (Data Lakes & Warehouses)
→ Latency in Streaming.
→ Error handling & logging. (Fault tolerance)

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1) Data Volume & Velocity
* Understands the scale of incoming data.
* High volume sources require scalable tools (Kafka, Kinesis)

2) Data Variety & Schema Management
* Handle structured, semi-structured & unstructured data.
* Maintain schema registry for consistent structure.

3) Data Quality & Validation
* Validate data before storage to prevent downstream errors.

4) Scalability & Performance
* Design ingestion pipelines to scale horizontally.

5) Reliability & Fault Tolerance
* Ensure ingestion systems can recover from failures.

6) Security & Compliance
* Encrypt data in transit and at rest.

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7) Metadata Management
* Track data sources, timestamps and transformations.

8) Monitoring & Observability
* Continuously monitori ingestion pipelines.

9) Cost optimization.

Batch Ingestion
* Batch Ingestion collects & processes data in groups or sets.

Workflow
1) Data Extraction from source systems.
2) Data Transformation & cleaning.
3) Data loading into target systems.
4) Scheduling & automation for repeated ingestion cycles.

Use cases
→ BI & reporting.
→ Data warehousing.
→ Historical data archiving.

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→ Data Ingestion is the bridge between data generation & data transformation.

Ways to Ingest Data

1) Direct Database Connections.

* Pulling data directly from databases, using standard connection interfaces primarily ODBC (Open Database Connectivity) & JDBC (Java Database Connectivity).

2) Change Data Capture

* This identifies & captures changes (INSERT, UPDATE, DELETE) from source databases in real time.

3) APIs for Ingestion

* These provide programmatic access to data from web applications, SaaS platforms & third-party systems.

→ APIs act as bridges between data producers (applications) & data consumers (data pipelines).

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4) Message Queues & Event Streaming Platforms

* To handle real-time flow, organizations use Message Queues and Event streaming Platforms

* These are middleware components that decouple producers & consumers allowing asynchronous, scalable, and reliable data ingestion.

Data Modeling

* A data model is a conceptual representation of how data is structured, organized, & related within a system.

→ Captures Business Logic.
→ Enables Data Consistency.
→ Connects Technical Design with Business Understanding.

Schema Level

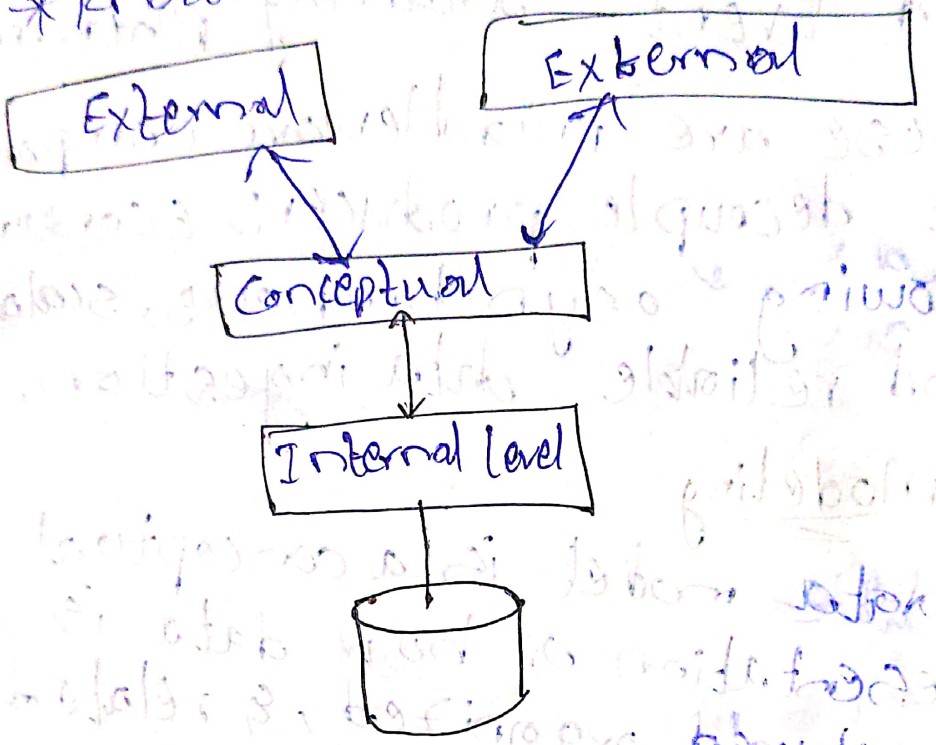
1) Internal

*Very low-level representation of entire database.
*Deals with physical records or blocks of database.

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Conceptual Level
* Representation of the entire Database

External Level
* known as user view level.



Data Grain
* Refers to the level of detail or resolution captured in a dataset.
* It defines what one row of data represents.

Normalization
* Organizing data in a database to reduce redundancy & improve data integrity.

→ 1NF
→ 2NF
→ 3NF

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Schema Evolution

* Ability of a system to handle changes in data structure over time, such as adding, renaming or removing fields in a stream.

Dimensional Modeling

* Used in Data Warehousing to organize data in a simple structure for easy querying, reporting and analysis.

Fact Table

* Stores numerical or measurable data.

Dimension Table

* Stores descriptive information about the facts.

Data Vault Modeling

* Used in data warehouse to store, integrate & track large amounts of data from multiple sources.

→ Hub = Stores unique business keys that identify entities.

→ Link = represents the relationship btwn two or more business entities.

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Data Transformation

* Converting raw data into from its original format into a clean, structured, & usable format to support analytics, M.L, reporting or application use.

Batch Transformations

* process discrete chunks of data on a scheduled basis.

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